ECG Signals for Heartbeat Classification

CS 539 - Team 25

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Problem & Motivation

- This project aims to design, implement, and evaluate various artificial neural network architectures for classifying ECG heartbeat signals using the MIT-BIH Arrhythmia dataset.

- Neural networks offer different strengths:
  - FNNs learn global patterns
  - CNNs extract local waveform features
  - LSTMs (RNNs) capture temporal dependencies
  - Hybrid CNN-LSTM models combine the best of both

- **Goal**: Compare FNN, CNN, LSTM, and CNN-LSTM models to identify the most robust approach for classifying ECG heartbeats from the MIT-BIH Arrhythmia dataset.

Project Scope & Goals

Scope

- Build, train, and evaluate multiple deep learning models for ECG heartbeat classification
  - **FNN**, **CNN**, **LSTM**, and **Hybrid CNN-LSTM** architectures
- Use the **MIT-BIH Arrhythmia–based ECG Heartbeat Dataset**
- Address dataset challenges:
  - **Class imbalance**
  - **Temporal patterns**
  - **Morphological variation** in heartbeat waveforms

Goals

- Compare model families to understand how **network complexity** affects performance
- Identify which architecture best captures:
  - **Morphological features** of ECG (CNN)
  - **Temporal dependencies** (LSTM)
  - **Combined sequence + feature representation** (CNN-LSTM)

Performance Metrics

- **Accuracy**
- **Precision, Recall, F1-Score (Macro & Weighted)**
- **Confusion Matrix**

Data Overview

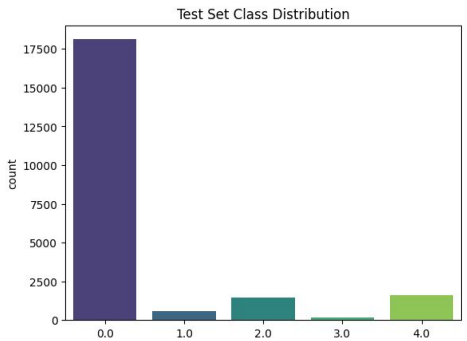
We use the ECG Heartbeat Categorization Dataset by Shayan Fazeli on Kaggle,which originates from the MIT-BIH Arrhythmia Database.

- Data Source: Physionet's MIT-BIH Arrhythmia Dataset
- Number of Classes/Categories: 5
- Classes: ['N': 0, 'S': 1, 'V': 2, 'F': 3, 'Q': 4]

Class	Samples	% of Test Dataset
0	18118	83%
1	556	3%
2	1448	7%
3	162	1%
4	1608	7%

Class Imbalance:

- EDA revealed a significant class imbalance in the dataset.
- Class 0 (normal beats) accounts for 80% of all training samples, whereas minority classes, particularly Class 3, represents less than 1% of the data.
- This imbalance leads to bias in model prediction; masking poor performance on clinically meaningful arrhythmia.



Methodology and Results

We implemented and compared four neural architectures for ECG heartbeat classification using the MIT-BIH–derived dataset:

1. Feedforward Neural Network (FNN)

- Serves as the baseline model.
- Fully connected layers learn global waveform relationships.
- Helps establish whether deep sequence/spatial models are necessary.

```
[FNN] Confusion Matrix:
[[16164  1225   262   352   115]
 [   65   470    15     2     4]
 [   29    13  1353    48     5]
 [    5     3     7   147     0]
 [    6    11     9     2  1580]]
```

2. 1D Convolutional Neural Network (CNN)

- Learns local ECG morphology (P-wave, QRS complex, T-wave).
- Convolution + pooling layers extract robust spatial features.
- Expected to outperform FNN due to better feature extraction.

```
[CNN] Confusion Matrix:
[[17506   210   116   262    24]
 [   78   461    12     4     1]
 [   18     7  1358    60     5]
 [    5     1     2   154     0]
 [   11     0     4     1  1592]]
```

Methodology and Results

3. Long Short-Term Memory Network (LSTM)

- Models the temporal dynamics of the ECG beat.
- Captures sequential dependencies using recurrent memory units.
- Useful for rhythms where beat shape evolves over time.

4. Hybrid CNN-LSTM Model

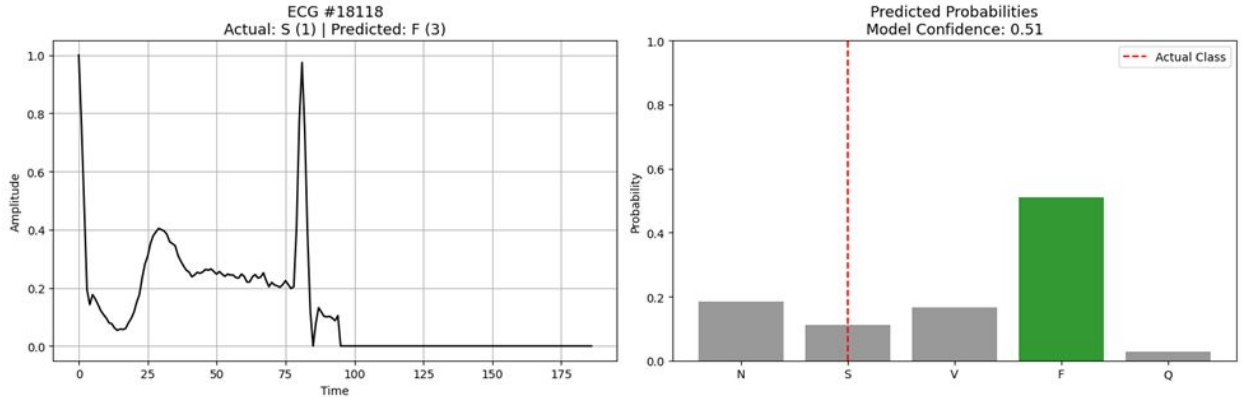
- CNN extracts local morphological patterns.
- LSTM models the temporal progression of extracted features.
- Expected to achieve the strongest performance by combining spatial + temporal learning.

```
[LSTM] Confusion Matrix:
[[17856    76   115    45    26]
 [  135   405    11     2     3]
 [   27     3  1379    36     3]
 [   17     0    10   134     1]
 [   19     3    15     2  1569]]

Confusion Matrix (Test):
[[17355   306   117   232   108]
 [   74   470     7     2     3]
 [   19     6  1354    58    11]
 [    3     0     7   152     0]
 [    7     3     3     2  1593]]
```

Results

Model	Accuracy	Class 0			Class 1			Class 2			Class 3			Class 4		
		Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score	Precision	Recall	F1 Score
FNN	0.9005	0.9935	0.8922	0.9401	0.2729	0.8453	0.4126	0.822	0.9344	0.8746	0.2668	0.9074	0.4123	0.9272	0.9826	0.9541
CNN	0.9625	0.9936	0.9662	0.9797	0.6789	0.8291	0.7466	0.9102	0.9378	0.9238	0.3202	0.9506	0.479	0.9815	0.99	0.9858
LSTM	**0.9749**	0.989	0.9855	0.9873	0.8316	0.7284	0.7766	0.9013	0.9523	0.9261	0.6119	0.8272	0.7034	0.9794	0.9757	0.9776
Hybrid (CNN+LSTM)	0.9429	0.9948	0.9403	0.9668	0.4084	0.8705	0.556	0.8786	0.9599	0.9175	0.441	0.8765	0.5868	0.9475	0.9882	0.9674



Sample Misclassification from the Hybrid (CNN + LSTM Model)

Discussion & Key Insights

- Our results show that the simple LSTM outperformed the hybrid CNN-LSTM, despite being less complex.
- Future work should investigate why deeper or hybrid models do not always yield better performance on ECG signals
- Implement Transformers (self-attention for long-range dependencies)
- Investigate synthetic data generation (SMOTE, GAN-based ECG synthesis)